

GET and POST Requests

FastAPI Masterclass

HTTP

- The **HyperText Transfer Protocol** (HTTP) is a web standard, a set of rules for how clients and servers send and receive data.
- A HTTP request includes a route and an **HTTP verb**, also called an **HTTP method**.
- The HTTP method describes the *type* of request.

GET Requests

- The **GET** HTTP method/verb indicates a request to fetch data.
- The server responds to a **GET** request by returning one or more records.
- The server may raise an error if the requested resource(s) cannot be found.

POST Requests

- The **POST** HTTP method indicates a request to create a record.
- For example, a **POST /orders** endpoint may request to create a new order record.
- The **POST** request will likely include the payload, the data that will form the record.

Status Codes

- The server response includes a **status code**, a numeric identifier that summarizes how the request went.
- For a **GET** request, the server returns **200 OK** if the requested resource was found and returned successfully.
- For a **POST** request, the server returns **201 Created** if the resource was created successfully.

Client Errors

- Something may go wrong with the request.
- Status codes in the 400 family represent **client errors**.
- For example, a **404 Not Found** status code indicates the client requested a resource that did not exist.
- A **401 Unauthorized** means the client is not authorized to access the requested resource.

Server Errors

- Status codes in the 500 family represent **server errors**.
- For example, a **500 Internal Server Error** status code indicates an error on the server side.
- As a simple example, a typo in a function may cause the route handler to raise an exception. FastAPI returns a 500.